

Answer Key & Teacher Notes

Session 1 – Equivalent Ratios and Tables • Session 2 – The Bridge Strategy (Scaling Ratios)

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Equivalent Ratios and Tables

TALK PROMPTS

- How can you tell whether every column in a ratio table shows the same ratio?
- Why does adding the same number to both rows change the ratio, but multiplying does not?

PARTNER TASK

Lemonade Recipe. One partner extends the 3 : 2 table by multiplying; the other checks each column by dividing back to 3 : 2. Switch roles with the 4 : 3 pancake batter table.

WATCH FOR

- Adding the same number to both terms (writing 4 : 6 as 6 : 8) instead of multiplying.
- Multiplying only one row of the table by the scale factor.
- Finding the factor from the wrong row (dividing the new milk amount by the old mix amount).

SESSION 2 • TEACHER NOTES

The Bridge Strategy (Scaling Ratios)

TALK PROMPTS

- Why is this called the “Bridge Strategy”?
- Why can simplifying help you reach 20 blue tiles from 24 without a whole-number multiplier?

PARTNER TASK

Salad Dressing, Concrete Mix, Candle Scents. Partners take turns: one builds the bridge ratio and names the divisor; the other scales the bridge to the target and checks by dividing back to the bridge.

WATCH FOR

- Adding the same amount to both terms to reach the target (writing 4 : 6 as 10 : 12).
- Multiplying by a factor that only gets “close” to the target ($12 \times 2 = 24$ instead of 20).
- Scaling only one term of the bridge ratio.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.RP.A.3a	1	1	1	2	1	2	2	3	3	1	1
S1 • Form B	6.RP.A.3a	1	1	1	2	1	2	2	3	3	1	1
S2 • Form A	6.RP.A.3a	1	1	2	2	2	2	2	3	3	1	2
S2 • Form B	6.RP.A.3a	1	1	2	2	2	2	2	3	3	1	2

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

equivalent ratios: Ratios that make the same comparison.

ratio table: A table of **equivalent** ratios. Each column is one pair.

scale factor: The number you **multiply** or divide both terms by.

- To find an equivalent ratio, multiply or divide **both** terms by the **same** number.
- Adding the same number to both terms does **not** keep the ratio.

Example: Find the scale factor: $15 \div 5 = 3$. Multiply the water by the same factor: $2 \times 3 = 6$. Check: $15 : 6$ divided by 3 gives **5 : 2** again. Answer: 15 scoops of powder go with **6** cups of water.

YOU TRY

- Water: 4, 8, 12, 20
- a. 8 b. 3

EXIT TICKET

- a. 15 b. 3
- White: 4, 8, 12
- B** (2 : 3)

PRACTICE FORM A

- a. 5 b. 6.4 c. 8 d. 15 e. 4 f. 5.5
- Syrup: 2, 4, 6, 8; 18 lemons: $18 \div 3 = 6$, $2 \times 6 = 12$ cups
- Mix: 4, 8, 12, 20; Milk: 3, 6, 9, 15
- Notebooks: 18, 9, 6, 3; Students: 6, 3, 2, 1
- $45 \div 15 = 3$, so $7 \times 3 = 21$ cups of sugar
- $27 \div 4.5 = 6$, so $9 \times 6 = 54$ pens
- $4.5 \div 1.5 = 3$, so $13.5 \times 3 = 40.5$ miles
- a. Sample: $5 : 12$ and $10 : 24$ b. No. $4.5 : 8$ adds 2 to each term. $2.5 \times 1.8 = 4.5$, but $6 \times 1.8 = 10.8$, not 8.
- Adding 2 to both terms does not keep the ratio; you must multiply or divide both by the same number. $4 : 6 \times 2 = 8 : 12$ (or $4 : 6 \div 2 = 2 : 3$).
- B** (8 : 20)
- C** (multiply both rows by the same number)

Self-check word **FACTOR**

PRACTICE FORM B

- a. 7 b. 6.3 c. 8 d. 28 e. 5 f. 6.5
- Syrup: 3, 6, 9, 12; 20 lemons: $20 \div 4 = 5$, $3 \times 5 = 15$ cups
- Oats: 5, 10, 15, 25; Honey: 2, 4, 6, 10
- Books: 36, 12, 8, 4; Carts: 9, 3, 2, 1
- $35 \div 7 = 5$, so $4 \times 5 = 20$ cups of raisins
- $21 \div 3.5 = 6$, so $7 \times 6 = 42$ crayons
- $7.5 \div 2.5 = 3$, so $12.5 \times 3 = 37.5$ miles
- a. Sample: $3 : 8$ and $6 : 16$ b. No. $3.5 : 6$ adds 2 to each term. $1.5 : 4 = 3 : 8$, but $3.5 : 6 = 7 : 12$, which is a different ratio.
- Adding 3 to both terms does not keep the ratio; you must multiply or divide both by the same number. $3 : 5 \times 2 = 6 : 10$.
- C** (9 : 12)
- B** (multiplied by 3)

Self-check word **SCALED**

GUIDED NOTES

bridge ratio: A **smaller** equivalent ratio you build first, then scale up.

simplify: Divide both terms by the **same** number.

greatest common factor: The **largest** number that divides both terms.

- When no whole number takes you to the target, **divide** both terms down to a bridge ratio.
- Then **multiply** both terms of the bridge up to the target.

Example: $6 \times$ a whole number never equals **15**. Divide both terms by 2: $6 : 8$ becomes **$3 : 4$** . Multiply both terms by 5: $3 : 4$ becomes $15 : 20$. Answer: 15 cups of nuts go with **20** cups of raisins.

YOU TRY

1. $8 : 20 \div 4 = 2 : 5$; $\times 7 = 14 : 35$, so 35 spoons of water
2. a. 15 b. 12

EXIT TICKET

1. a. $3 : 5$ b. $2 : 3$
2. $6 : 4 \div 2 = 3 : 2$; $\times 5 = 15 : 10$, so 15 cups
3. **B** (divide both terms by the same number)

PRACTICE FORM A

1. a. $2 : 3$ b. $3 : 4$ c. $4 : 5$ d. $4 : 9$
2. Bridge $2 : 3 (\div 2)$; $10 : 15 (\times 5)$, so 15 cups of yellow
3. a. $6 : 9 \div 3 = 2 : 3$ b. $2 : 3 \times 7 = 14 : 21$, so 21 tbsp
4. $9 : 15 \div 3 = 3 : 5$; $\times 7 = 21 : 35$, so 35 bags
5. $12 : 18 \div 6 = 2 : 3$; $\times 5 = 10 : 15$, so 15 drops
6. $24 : 30 \div 6 = 4 : 5$; $\times 5 = 20 : 25$, so 25 cream tiles
7. a. $12 \div 5 = 2.4$ L b. $2.4 \times 18 = 43.2$ L
8. a. $10 \div 4 = 2.5$ cups b. $2.5 \times 18 = 45$ cups c. $18 \div 4 = 4.5$ is not a whole number, so bridge through 1 smoothie (or $5 : 2$) instead.
9. Adding 6 to both terms does not keep the ratio. Bridge: $4 : 6 \div 2 = 2 : 3$, then $\times 5 = 10 : 15$. He needs 15 cups of yellow.
10. **D** (3)
11. **B** (15)

Self-check word BRIDGES

PRACTICE FORM B

1. a. $3 : 5$ b. $2 : 3$ c. $3 : 4$ d. $4 : 9$
2. Bridge $3 : 5 (\div 2)$; $9 : 15 (\times 3)$, so 15 cups of yellow
3. a. $8 : 12 \div 4 = 2 : 3$ b. $2 : 3 \times 10 = 20 : 30$, so 30 cups
4. $21 : 28 \div 7 = 3 : 4$; $\times 5 = 15 : 20$, so 20 oz
5. $10 : 15 \div 5 = 2 : 3$; $\times 8 = 16 : 24$, so 24 white beads
6. $16 : 36 \div 4 = 4 : 9$; $\times 3 = 12 : 27$, so 27 cups
7. a. $15 \div 4 = 3.75$ L b. $3.75 \times 10 = 37.5$ L
8. a. $15 \div 6 = 2.5$ cups b. $2.5 \times 14 = 35$ cups c. $14 \div 6$ is not a whole number, so bridge through 1 smoothie (or $5 : 2$) instead.
9. Adding 2 to both terms does not keep the ratio. Bridge: $8 : 12 \div 4 = 2 : 3$, then $\times 5 = 10 : 15$. She needs 15 cups of white.
10. **B** (4.5)
11. **C** (25)

Self-check word SCALING

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • EQUIVALENT RATIOS AND TABLES

1. Nuts: 2, 4, 6, 10, 14; Cereal: 5, 10, 15, 25, 35
2. a. 20 b. 36 c. 4 d. 14
3. A, B, D ($4 : 6 \div 2 = 2 : 3$; $\times 2 = 8 : 12$; $2 : 3 \times 5 = 10 : 15$; 6 : 8 adds 2 to each term)
4. $9 \times 4 = 36$, so $4 \times 4 = 16$ gold beads (table: 4 : 9, 8 : 18, 12 : 27, 16 : 36)
5. a. $3 \times 7 = 21$, so $2 \times 7 = 14$ cups of white b. $21 + 14 = 35$ cups
6. No. Adding 3 to both terms does not keep the ratio; you must multiply or divide both terms by the same number. Example: $5 : 8 = 10 : 16 (\times 2)$.

SHEET 2 • THE BRIDGE STRATEGY (SCALING RATIOS)

1. 2 minutes: 5 bottles ($\div 3$); 10 minutes: 25 bottles ($\times 5$)
2. a. 15 b. 8
3. a. $4 : 30 \div 2 = 2 : 15$ b. $2 : 15 \times 5 = 10 : 75$, so 75 km
4. a. $10 : 4 \div 2 = 5 : 2$ b. $5 : 2 \times 3 = 15 : 6$, so 15 cups
5. Adding 9 to both terms does not keep the ratio.
Bridge: $6 : 10 \div 2 = 3 : 5$, then $\times 5 = 15 : 25$. She needs 25 spoons of water.
6. **A** (10)
7. **A** ($\div 3$, then $\times 5$)